

Which of the following statements are true concerning the molar mass of ammonium carbonate?

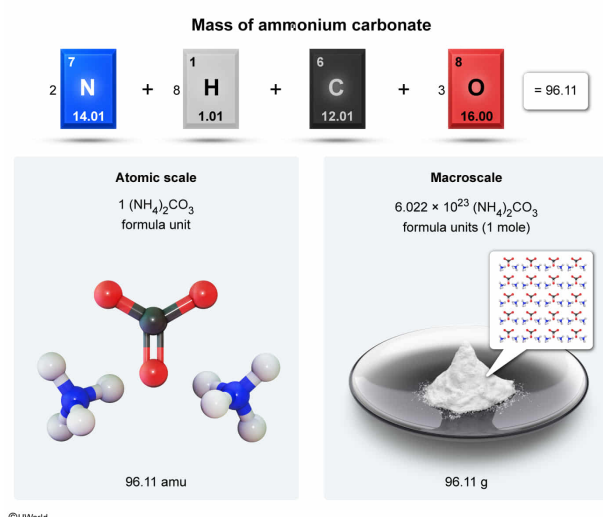
- I. 1.00 mol of $(\text{NH}_4)_2\text{CO}_3$ contains 96.1 g of mass.
- II. 96.1 g of $(\text{NH}_4)_2\text{CO}_3$ contains 6.02×10^{23} $(\text{NH}_4)_2\text{CO}_3$ formula units.
- III. 96.1 amu of $(\text{NH}_4)_2\text{CO}_3$ contains 1.00 mol $(\text{NH}_4)_2\text{CO}_3$.

- ✓ ☒ A. I and II only [47%]
☐ B. I and III only [15%]
☐ C. II and III only [3%]
☐ D. I, II, and III [32%]

Correct

46% answered correctly

Explanation:



An **atomic mass unit** (amu) is defined as one-twelfth the mass contained in a carbon-12 atom (ie, the average mass of 6 protons and 6 neutrons). Because all atoms are made of protons and neutrons, the amu makes a convenient unit to assess the **atomic masses of other elements** on the periodic table.

The mass of a **formula unit** of an ionic compound can also be expressed in amu by calculating the sum of the **atomic masses** of each atom in the formula unit. For $(\text{NH}_4)_2\text{CO}_3$, the mass of the formula unit is given by:

$$\text{Mass}(\text{NH}_4)_2\text{CO}_3 = \text{Mass}(2 \text{ N} + 8 \text{ H} + 1 \text{ C} + 3 \text{ O})$$

Substituting the mass values listed on the **periodic table** gives:

$$\text{Mass } (\text{NH}_4)_2\text{CO}_3 = 2(14.0 \text{ amu}) + 8(1.0 \text{ amu}) + 12.0 \text{ amu} + 3(16.0 \text{ amu}) = 96.1 \text{ amu}$$

To relate atomic-scale masses (amu) of an individual formula unit to macroscale masses (grams) of samples containing trillions of formula units, a counting unit called the **mole** (abbreviated "mol") is defined as the **number of carbon-12 atoms** required to give a mass of exactly 12.00 grams. This number of atoms is a constant called Avogadro's number ($N_A = 6.02 \times 10^{23}$).

$$1 \text{ mole} = 6.02 \times 10^{23} \text{ units}$$

$$6.02 \times 10^{23} \text{ atoms of } ^{12}\text{C} = 1 \text{ mole } ^{12}\text{C} = 12.0 \text{ g } ^{12}\text{C}$$

Because each carbon-12 atom contains 12 amu, the mole can also be viewed more generally as the number of amu required to give exactly 1.00 gram of mass.

$$6.02 \times 10^{23} \text{ amu} = 1 \text{ mole of amu} = 1.00 \text{ g}$$

Consequently, the mass of a single molecule or formula unit (in amu) numerically matches its **molar mass** (in g/mol), but the quantities have different units to indicate the scale (ie, atomic scale versus macroscale).

Therefore, a molecular weight of 96.1 amu for $(\text{NH}_4)_2\text{CO}_3$ means that 1.00 mole of $(\text{NH}_4)_2\text{CO}_3$ molecules has a mass of 96.1 g (**Number I**). Likewise, a 96.1 g sample of $(\text{NH}_4)_2\text{CO}_3$ contains 1 mole (6.02×10^{23} units) of $(\text{NH}_4)_2\text{CO}_3$ formula units (**Number II**).

(**Number III**) The **mass of a single $(\text{NH}_4)_2\text{CO}_3$ unit** is 96.1 amu, but 1.00 *mole* of $(\text{NH}_4)_2\text{CO}_3$ formula units would have 6.02×10^{23} times this number of amu.

$$1.00 \text{ mole } (\text{NH}_4)_2\text{CO}_3 = 6.02 \times 10^{23} \text{ units of } (\text{NH}_4)_2\text{CO}_3$$

$$6.02 \times 10^{23} \text{ units of } (\text{NH}_4)_2\text{CO}_3 \times \frac{96.1 \text{ amu}}{1 \text{ units of } (\text{NH}_4)_2\text{CO}_3} = 5.79 \times 10^{25} \text{ amu}$$

In other words, 1.00 mol of $(\text{NH}_4)_2\text{CO}_3$ contains 5.79×10^{25} amu of $(\text{NH}_4)_2\text{CO}_3$, not 96.1 amu of $(\text{NH}_4)_2\text{CO}_3$.

Educational objective:

The molecular weight is the sum of the atomic mass units contributed from each atom in a formula unit. One mole (6.02×10^{23} formula units) of a given molecular mass equals the molar mass, which is the amount of mass (in grams) contained in 1 mole of the formula units (grams/mole).